Yeruru Asrar Ahmed | Lead Engineer | Generative AI | Computer Vision

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Professional Summary

Lead Engineer at **Samsung Research Institute Bangalore**, driving innovations in **multi-modal generation, federated learning, and on-device personalization**. Ph.D. from **IIT Madras** specializing in **text-to-image synthesis** and **representation learning**. Research interests span generative vision AI and scalable AI systems.

Core Research and Technical Interests

Generative AI, Multi-modal Learning, Diffusion Models, Representation Learning, Federated Learning, Vision-Language Models, Large Multi-Modal Models (LMMs), PEFT, and AI Systems Optimization.

Professional Experience

Samsung Research Institute Bangalore
Lead Engineer — *May 2025–Present*

- **Platform Intelligence Team:** Working on on-device personalization and Federated Learning (FL) systems for Samsung S26.
- **Nudge Action (S26):** Built and trained a LoRA-based classification model for action prediction.
- **Personalized Data Engine:** Modelled system usage patterns to enable intelligent background task scheduling in S26 (logging for analysis).
- **Federated Learning Stabilization:** Improved convergence stability and researched on handling client heterogeneity.
- **One-Shot FL:** Developed and evaluated one-shot federated learning approaches for domain adaptation in on-device image personalization.
- **Current Research:** PEFT-based multi-task optimization to reduce memory footprint across multiple LoRA-based models.

8bit.ai (CtrlS–Cloud4C)
AI Scientist — *Dec 2024–Mar 2025*

- Worked on agentic AI systems for enterprise analytics platform Neutrino.
- Designed computer vision pipelines for real-time safety monitoring and compliance tracking.
- Collaborated with manufacturing and finance clients to deploy scalable AI solutions.

Sony Research India
Research Intern — *Nov 2023–Aug 2024*

- Worked on text-to-video distillation to accelerate diffusion models while maintaining visual fidelity.
- Leveraged Latent Consistency Models (LCM) and PEFT for parameter-efficient training.
- Applied LLM-guided caption refinement to improve grounding and video-text coherence.

Publications

- **Unsupervised Co-generation of Foreground–Background Segmentation from Text-to-Image Synthesis**, WACV 2024; extended version in CVIU 2024.
- **End-to-end Training for Text-to-Image Synthesis using Dual-Text Embeddings**, TMLR 2025.
- **Self-Distillation on Conditional Spatial Activation Maps for FG–BG Segmentation**, CVPRW 2024.
- **Breast Cancer Segmentation using UNet and Global Convolutional Networks**, ICPR 2024.
- **Twitter Sentiment Visualization using Deep Learning**, ICIoTCT 2020.

Education

Indian Institute of Technology Madras — **CGPA: 8.5/10**
Ph.D. in Computer Science and Engineering — *2018–2025*
Thesis: *Text-to-Image Synthesis with Novel Representations and Corollaries*

National Institute of Technology Hamirpur — **CGPA: 9.42/10**
M.Tech in Computer Science — *2016–2018*

GITAM University, Visakhapatnam **CGPA: 8.5/10**
B.Tech in Computer Science *2012–2016*

Selected Research Projects

- **COS-GAN:** Co-generation of images and segmentation masks using spatial co-attention; demonstrated unsupervised learning for image synthesis and segmentation.
- **Dual-Text Embedding GAN:** Jointly optimized multi-level text conditioning for semantically accurate image generation.
- **Self-Distillation Maps:** Foreground–background segmentation via single-step distillation and reconstruction networks.
- **Residual Multi-Scale CNN:** Dense atrous convolutional model for semantic segmentation on PASCAL VOC.

Academic and Teaching Experience

- Teaching Assistant at IIT Madras (2018–2023) for courses including Deep Learning, Pattern Recognition and Machine Learning, Operating Systems, and Computer Architecture.
- Mentored 20+ undergraduate projects in computer vision, GAN-based synthesis, and deep learning systems.

Awards and Achievements

- **Star Teaching Assistant** Award (3×) – Department of Computer Science, IIT Madras.
- Secured **GATE Rank 2674 (2017)** and **3612 (2016)** among 100,000+ candidates.
- Represented university in **Inter-University Football Tournament, South Zone, Calicut University, 2014**.

Technical Skills

- **Programming:** Python, C, C++
- **Frameworks:** PyTorch, TensorFlow, OpenCV, HuggingFace Transformers
- **Expertise:** Diffusion Models, GANs, Vision-Language Models, Federated Learning, Edge AI, LLM Integration

Additional Information

- **Hobbies:** Football, Cooking, Movies
- **Languages:** English, Hindi, Telugu

Declaration

I hereby declare that the information furnished above is true to the best of my knowledge and belief.

Place: Bengaluru, Karnataka **Yeruru Asrar Ahmed**
Date: March 1, 2026